

Search Engine using BM25 and SBERT

ECS736P – Information Retrieval

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Objective

- Build a search engine for financial queries
- Retrieve and rank relevant documents
- Compare lexical (BM25) and semantic (SBERT) retrieval
- Evaluate performance using IR metrics

Dataset

- Dataset: FiQA-2018 (BEIR benchmark)
- Financial question-answering dataset
- Test split used for evaluation
- 57,638 documents (corpus)
- 648 queries
- 648 relevance judgements (qrels)

System Pipeline

- Dataset loading
- BM25 indexing (lexical baseline)
- BM25 parameter tuning (k_1 , b)
- SBERT dense retrieval (semantic)
- Evaluation and comparison

Dataset → BM25 → Tuning → SBERT → Evaluation

BM25 (Lexical Retrieval)

- Probabilistic lexical retrieval model based on keyword matching
- Scores documents using:
 - term frequency
 - inverse document frequency
 - document length
- Efficient and interpretable
- Strong baseline for information retrieval

BM25 Scoring Function

$$\text{BM25}(q, d) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{f(t, d) \cdot (k_1 + 1)}{f(t, d) + k_1 \left(1 - b + b \cdot \frac{|d|}{\text{avgdl}} \right)}$$

- Combines term frequency, document length, and term importance

SBERT (Semantic Retrieval)

- Uses sentence embeddings for semantic similarity
- Captures meaning beyond exact word matching
- Implemented using FAISS for efficient search
- Improves retrieval when wording differs

Evaluation Metrics

- $\text{NDCG}@10$ → ranking quality (primary metric)
- $\text{Precision}@10$ → relevance in top results
- $\text{Recall}@100$ → coverage of relevant documents
- $\text{MAP}@100$ → overall performance across queries

Results

- **BM25 Baseline**

- NDCG@10: 0.2345

- **BM25 Tuned**

- NDCG@10: 0.2401
- Best parameters: $k1 = 0.9$, $b = 0.75$

- **SBERT (Semantic Retrieval)**

- NDCG@10: 0.4443

- SBERT significantly improves retrieval performance
- SBERT improves NDCG@10 by ~85% over baseline
0.2345 → 0.4443

Conclusion

- BM25 provides a strong baseline for retrieval
- Parameter tuning gives small improvements
- SBERT significantly improves retrieval performance
- Semantic retrieval handles vocabulary mismatch
- System demonstrates effective real-world search performance